Low-Complexity Control for Vision-Based Landing of Quadrotor UAV on Unknown Moving Platform

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***Abstract*—This article addresses the vision-based landing problem of a low-cost quadrotor on an unknown moving platform. A robust landing controller is developed, which consists of the design of the low-complexity outer-loop controller and the geometric inner-loop attitude controller. First, an error transformation based on prescribed performance is designed to guarantee the landing behaviors and deal with the intermediate control signal of backstepping approaches, resulting in a low-complexity position-based visual servoing (PBVS) design. In addition, the proposed PBVS controller exhibits strong robustness against an uncertain relative dynamic system due to no incorporation of any prior knowledge of the moving platform. Next, a modified geometric attitude controller is presented by characterizing the geometric properties of rotation matrices intrinsically. Finally, the stability analysis is presented using Lyapunov stability theory, and the effectiveness of the proposed control strategy is demonstrated through numerical simulations and experiments.**

***Index Terms*—Autonomous landing, dynamic position-based visual servo, low-complexity, prescribed performance, unmanned aerial vehicle (UAV).**

I. Introduction

DUE to the vertical takeoff and landing (VTOL) as well as hovering capabilities, the past few decades have witnessed a growing interest in VTOL unmanned aerial vehicles (UAVs). Particularly, quadrotors, with the features of low cost and small size, serve as the ideal aerial platforms that have been widely employed in military and civilian applications, such as border surveillance, disaster monitoring, and search and rescue missions [1], [2]. The landing of the quadrotor is a fundamental maneuver, but it remains a delicate process during the flight [3], since the landing control design suffers from both the limited sensing of the quadrotors and the unknown model of the moving targets. As a result, the landing control of the low-cost quadrotor on the unknown moving target is an extremely challenging and valuable problem.

Sensing the environment to locate itself is a prerequisite for the quadrotor autonomous landing. In general, the global pose can be provided by a global positioning system (GPS), however, the GPS signals degrade severely indoors or in clustering environments such as cities and forests. Alternatively, the onboard sensors provide robust relative pose information and allow more general autonomous navigation. Thus, some solutions have been explored based on onboard sensors in the GPS-denied environments, including sonar, laser range finder, LIDAR (light detection and ranging), and visual cameras. Among them, the visual cameras are of low cost and lightweight that can provide richer information [4], making them a very popular option for landing control of quadrotors.

Various vision-based landing control approaches have been developed for the quadrotor UAVs, and they can be divided into two basic types: image-based visual servo (IBVS) [5]–[7], and position-based visual servo (PBVS) [8]–[12]. In IBVS, the image features are directly used in the formulation of the control laws. Nevertheless, it may produce partly undesired motion trajectories in the Cartesian space, especially when it needs a large rotation around the optical axis of the camera. The PBVS approach requires the explicit calculation of the pose estimation based on the image features, but at the same time, it separates the controller design problem from the vision-based pose estimation problem and leads to a straightforward control design problem in the Cartesian space. Owing to the underactuated dynamics of the quadrotor UAVs, we are interested in the PBVS schemes that avoid the image Jacobian formulation. In this direction, several classical linear control methods have been proposed in [8]–[10]. Since almost all actual dynamic systems have the characteristics of uncertainty and nonlinearity, other relevant nonlinear

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controllers that employed nested saturations, backstepping, and sliding modes control techniques were presented in [11]. Also, a hybrid automaton was designed to change control modes for the landing maneuver in [12] to counteract the uncertainty model of the slope. However, all the aforementioned PBVS schemes were developed for a static landing platform.

Although the PBVS landing schemes on a static platform have achieved many results, for more general applications, the challenge of stabilizing the motion of the quadrotors relative to the moving platforms is greatly increased. Some interesting works about landing on a moving platform have been investigated. In [13], the velocity of the moving target was assumed to be known, but the fact that the motion of the platform is not usually available in a real environment. In [14], the PID controller has good performance for landing on a high-speed vehicle under the premise that the position and the velocity of the landing vehicle can be measured. In [15]–[20], several schemes based on the model of the landing platform have been developed. Usually, a vertical movement of the platform was modeled as a superposition of several sine functions with different amplitudes, frequencies, and phases [15]–[17], and a horizontal movement of the platform was modeled as a car-like model [18]–[20]. Along the same line, the Kalman filters [15]–[19] and velocity observer [20] were used for motion estimation of a moving platform. However, the estimation of the model parameters or velocity is rather tedious, and the controller design is conservative and only available in the presence of the moving platform models. For another, some robust control schemes were proposed in [21] and [22]. In [21], the invariant ellipsoid technique was proposed to reasonably compensate for the ship heave motion up to within a certain error bound. The neural network control method was used in [22] to improve robustness against the uncertainties of the moving platform, but it was difficult to implement on real-time control systems due to the complicated control algorithm.

The motivation of this article is to propose a low-complexity dynamic PBVS scheme for landing on an unknown moving platform. Aiming at the underactuated nature of the quadrotor, our control scheme is decomposed into an outer-loop position control and an inner-loop attitude control module. Motivated by the prescribed performance control schemes [23]–[25], the error transformation based on the performance function is designed to guarantee the landing behaviors and handle the uncertain nonlinear landing systems. Using the Lyapunov stability analysis, stability and thus prescribed performance of the autonomous landing system can be guaranteed. Compared with the existing PBVS landing control schemes, the main contributions of this article can be summarized as follows.

1) A novel low-complexity PBVS controller of the low-cost quadrotor is proposed for the landing on the unknown moving target. Particularly, the error transformation approach is designed to deal with the intermediate signal of the backstepping control technique. It completely avoids the explosion of the complexity issue raised by traditional backstepping-like approaches, without utilizing additional filters. Therefore, this method simplifies the control design and performs well in real-time uncertain dynamic systems. In contrast, some methods sacrificed computational resources to improve robustness against the uncertain relative dynamic systems. The method of the optimal control was used in [21], and the neural network controller was adopted in [22].
2) The landing controller does not incorporate any prior knowledge of the moving platforms, so it exhibits strong robustness against the uncertain relative dynamic system. Different from methods of cooperative landing [14] and known relative dynamic models [15]–[20], the proposed method is a noncooperative autonomous landing and it is isolated from prior model selections and velocity measurements of the moving platforms.
3) By using prescribed performance functions, the landing error is regulated to a predefined arbitrarily small residual set along with the absolute decaying time function, and the maximum overshoot is less than a prescribed value. Compared with landing solutions in [13]–[20], the proposed method can explicitly guarantee the transient and steady-state landing behaviors.

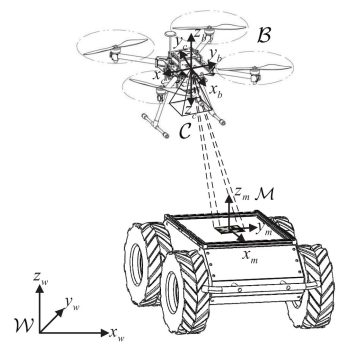

Fig. 1. Vision-based landing system and involved coordinate frames.

The rest of this article is organized as follows. Section II introduces system modeling and problem statements. Section III presents the control design and stability analysis. Simulation and experimental results to verify the effectiveness of the proposed control method are shown in Section IV. Finally, Section V concludes this article.

II. Problem Formulation

A. System Description and Modelling

Consider the landing system that includes a quadrotor UAV equipped with a vision suite and a fiducial landmark attached to a moving platform, as shown in Fig. 1. The inertial frame $\mathcal{W}$ is rigidly attached to a position on the earth in East-North-Up coordinates. The body-fixed frame $\mathcal{B}$ is defined to describe the motion equation of the quadrotor. Let $\mathcal{C}$ be the camera frame, and $\mathcal{M}$ be the landmark frame. Let $\boldsymbol{p} \in \mathbb{R}^3$ and $\boldsymbol{v} \in \mathbb{R}^3$ denote the position and linear velocity of quadrotor expressed in $\mathcal{W}$, $\boldsymbol{R} \in SO(3)$ is the rotation matrix from $\mathcal{B}$ to $\mathcal{W}$, and $\boldsymbol{\Omega} \in \mathbb{R}^3$ is the angular velocity expressed in $\mathcal{B}$. Using the Newton–Euler formulation, the dynamics of the quadrotor is given by

$$\dot{\boldsymbol{p}} = \boldsymbol{v} \tag{1}$$

$$m\dot{\boldsymbol{v}} = -mg\boldsymbol{i}_3 + f\boldsymbol{R}\boldsymbol{i}_3 \tag{2}$$

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$$\dot{\boldsymbol{R}} = \boldsymbol{R}\boldsymbol{\Omega}^{\wedge} \tag{3}$$

$$\boldsymbol{J}\dot{\boldsymbol{\Omega}} = -\boldsymbol{\Omega} \times \boldsymbol{J}\boldsymbol{\Omega} + \boldsymbol{\tau} \tag{4}$$

where m represents the mass; $\boldsymbol{J} = [J_x, J_y, J_z]^T$ is the inertia matrix of the quadrotor; g is the acceleration of gravity, and $\boldsymbol{i}_3 = [0, 0, 1]^T$ is a unit vector. The thrust force f and the torque vector $\boldsymbol{\tau} = [\tau_x, \tau_y, \tau_z]^T$ are expressed in the frame $\mathcal{B}$. The hat map $(\cdot)^{\wedge} : \mathbb{R}^3 \mapsto SO(3)$ denotes the skew-symmetric matrix such that $\boldsymbol{\Omega}^{\wedge}\boldsymbol{x} = \boldsymbol{\Omega} \times \boldsymbol{x}$ for any $\boldsymbol{x}$.

Let $\boldsymbol{p}_t \in \mathbb{R}^3$ expressed in $\mathcal{W}$ be the origin of the landmark frame $\mathcal{M}$ attached to the moving platform. Since the landing platform is moving, we denote the velocity of the platform as $\dot{\boldsymbol{p}}_t \in \mathbb{R}^3$. Instead of measuring and estimating the velocity of the platform, we only assume that:

Assumption 1: The velocity of the moving platform $\dot{\boldsymbol{p}}_t$ is unknown, but there exist an unknown real positive constant $\bar{\nu}_t$ such that $\|\dot{\boldsymbol{p}}_t\| < \bar{\nu}_t$.

B. Vision-Based Position Estimation

A perspective camera is attached to the quadrotor to estimate its position relative to the landmark during landing. According to the pin-hole model, the mapping from a point $\boldsymbol{P}^c = [X, Y, Z]^T$ in the camera frame to an image coordinate $\boldsymbol{s} = [u, v]^T$ is defined as

$$\boldsymbol{s} = \begin{bmatrix} u \\ v \end{bmatrix} = \frac{f_c}{Z} \begin{bmatrix} X \\ Y \end{bmatrix} \tag{5}$$

where $f_c > 0$ is the focal length of the camera. We consider a squared fiducial marker AprilTag [26] as a landmark to estimate the relative position. The 3-D position of the square corner of the AprilTag is denoted as $\boldsymbol{P}^m$ in $\mathcal{M}$, and the corresponding position $\boldsymbol{P}^c$ expressed in $\mathcal{C}$ is given by

$$\boldsymbol{P}^c = \boldsymbol{R}_M^C \boldsymbol{P}^m + \boldsymbol{t}_M^C \tag{6}$$

where $\boldsymbol{R}_M^C \in SO(3)$ and $\boldsymbol{t}_M^C \in \mathbb{R}^3$ represent the rotation and translation of $\mathcal{M}$ relative to $\mathcal{C}$, respectively. They can be estimated by using the four square corners of the AprilTag, and the side length of the AprilTag is the only scale information for the entire process. As a result, the position of the quadrotor relative to the center of the landmark expressed in $\mathcal{W}$ can be obtained as

$$\boldsymbol{r}_{pt} = \boldsymbol{p} - \boldsymbol{p}_t = -\boldsymbol{R}\boldsymbol{R}_C^B \boldsymbol{t}_M^C \tag{7}$$

where $\boldsymbol{R}_C^B$ is a known fixed rotational transformation of the camera frame relative to the quadrotor body frame.

C. Problem Statement

The objective of this article is to design the PBVS controller for a quadrotor landing on a moving platform with model uncertainty. The relative position between the quadrotor and the landmark is estimated by a monocular camera, and the linear/angular velocity, as well as attitude angles of a quadrotor, are provided by an auxiliary inertial measurement unit (IMU). The linear velocity $\dot{\boldsymbol{p}}_t$ and angular velocity ω_t of the landmark attached to the moving platform cannot be measured or estimated due to the uncertain model of the moving platform. The quadrotor maneuvers itself toward the safe landing point above the moving platform, and it performs vertical landing on the moving platform by turning OFF motors when the quadrotor is sufficiently close to the landing point. Specifically, given the desired relative position $\boldsymbol{r}_{pt}^*$, the goal is to design a controller for the quadrotor so that $\boldsymbol{r}_{pt}$ converges to a predefined and arbitrarily small neighborhood of $\boldsymbol{r}_{pt}^*$.

III. Control Design

A. Error Transformation

In this section, the error transformation for the position control is derived. In particular, it quantitatively characterizes the landing performance. For completeness and compactness of presentation, the preliminary knowledge on prescribed performance is first summarized. The prescribed performance means that the minimum speed of the convergence, the maximum steady-state error, and the maximum allowable overshoot are set *a priori*. Typically, the mathematical expression of the prescribed performance is given by the following inequalities:

$$-\rho_i(t) < e_i(t) < \rho_i(t), \ \forall t > 0 \tag{8}$$

where $\rho_i(t)$ is a continuously differentiable, bounded, strictly positive, and decreasing function of time, called performance function, which can be defined as

$$\rho_i(t) = (\rho_i(0) - \rho_i^{\infty})e^{-l_i t} + \rho_i^{\infty} \tag{9}$$

where $\rho_i(0) > 0$ is a positive parameter indicating the maximum allowable size of the tracking error, thus the initial error can be guaranteed as $0 \leq \|e_i(0)\| < \rho_i(0)$. The decreasing rate of $\rho_i(t)$ depends on the constant $l > 0$, and $\rho_i^{\infty} > 0$ represents the steady-state performance specification, which can be set small enough to meet performance requirements.

To achieve landing control with the prescribed performance, an error transformation is designed as follows:

$$e_i = T(\varepsilon_i)\rho_i(t) \tag{10}$$

where the error transformation function $T(\varepsilon_i)$ is a smooth strictly increasing function with respect to a new error ε_i, and the inverse function of $T(\varepsilon_i)$ for ε_i exists. Besides, $T(\varepsilon_i)$ satisfies the following properties:

1) $-1 < T(\varepsilon_i) < 1, \ \forall \varepsilon_i \in \mathbb{R}$
2) $\lim_{\varepsilon_i \to +\infty} T(\varepsilon_i) = 1$
3) $\lim_{\varepsilon_i \to -\infty} T(\varepsilon_i) = -1$.

Based on the above requirements, a hyperbolic-tangent-type function can be chosen as the candidate function $T(\varepsilon_k)$ as

$$T(\varepsilon_i) = \frac{e^{\varepsilon_i} - e^{-\varepsilon_i}}{e^{\varepsilon_i} + e^{-\varepsilon_i}}. \tag{11}$$

If we denote $\xi_i = e_i/\rho_i$, then the transformed error ε_i can be expressed via the inverse transformation of (11) as

$$\varepsilon_i = \frac{1}{2} \ln\left(\frac{1 + \xi_i}{1 - \xi_i}\right) \tag{12}$$

where $\ln(\cdot)$ denotes the natural logarithm. At this point, it can be seen that if $e_i \to 0$, then $\varepsilon_i \to 0$. Furthermore, the prescribed performance (8) will be achieved if the transformed error implemented in the controller design is bounded.

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Remark 1: The key idea of the error transformation is to transform the original nonlinear system with the constrained landing error behavior appearing in (8), into an equivalent unconstrained one. To this end, several common transformation functions, including the arctangent-type or the hyperbolic-tangent-type [23]–[25], are used. Owing to the property of these two kinds of functions, both of them are feasible to guarantee the domain of the inverse of the transformation function is a bounded open interval. As a result, we can achieve the prescribed performance constraint $|\xi_i| < 1$. Since the hyperbolic-tangent-type method is more popular, it is used in the proposed method. Therefore, the landing error strictly remains within the prescribed error bound as long as the stability of the transformed error system is ensured. In the end, the designed controller based on the transformed error can explicitly meet the requirements of transient and steady-state performance for landing in many applications of interest.

B. Low Complexity Position Control

To stabilize the translation dynamic subsystem (1) and (2) and guarantee the landing with prescribed performance, the position control scheme is proposed based on the aforementioned error transformation. Define the position error $\boldsymbol{e}_p$ as

$$\boldsymbol{e}_p = \boldsymbol{r}_{pt} - \boldsymbol{r}^*_{pt}. \tag{13}$$

Using (12), the position error can be transformed to a new error $\boldsymbol{\varepsilon}_p = [\varepsilon_{p1}, \varepsilon_{p2}, \varepsilon_{p3}]^T$, and its derivative is given by

$$\dot{\boldsymbol{\varepsilon}}_p = \boldsymbol{Q}_p(\boldsymbol{v} - \dot{\boldsymbol{p}}_t + \boldsymbol{\varrho}_p) \tag{14}$$

where $\boldsymbol{Q}_p = \mathrm{diag}(q_{p1}, q_{p2}, q_{p3})$; $\boldsymbol{\varrho}_p = [\varrho_{p1}\varrho_{p2}, \varrho_{p3}]^\mathsf{T}$; q_{pj} and ϱ_{pj} for $j \in 1, 2, 3$ are defined as

$$q_{pj} = \frac{\partial \varepsilon_{pj}}{\partial e_{pj}} = \frac{1}{(1+\xi_{pj})(1-\xi_{pj})}$$

$$\varrho_{pj} = \frac{\partial \xi_{pj}}{\partial t} = -\dot{\rho}_{pj}\xi_{pj}.$$

Following the backstepping control method, the intermediate virtual velocity variable $\hat{\boldsymbol{v}} = [\hat{v}_1, \hat{v}_2, \hat{v}_3]^T$ is introduced to guarantee the stability of $\boldsymbol{\varepsilon}_p$, and it is designed as

$$\hat{\boldsymbol{v}} = -k_1\boldsymbol{Q}_p^\mathsf{T}\boldsymbol{\varepsilon}_p \tag{15}$$

where $k_1 > 0$ is a positive gain.

Define a linear velocity error signal as

$$\boldsymbol{e}_v = \boldsymbol{v} - \hat{\boldsymbol{v}} \tag{16}$$

where $\boldsymbol{e}_v = [e_{v1}, e_{v2}, e_{v3}]^T$. Similarly, the linear velocity error can also be transformed to $\boldsymbol{\varepsilon}_v = [\varepsilon_{v1}, \varepsilon_{v2}, \varepsilon_{v3}]^T$. By recalling (2), the time derivative of transformed error $\boldsymbol{\varepsilon}_v$ is

$$\dot{\boldsymbol{\varepsilon}}_v = \boldsymbol{Q}_v(-g\boldsymbol{i}_3 + \frac{f\boldsymbol{R}\boldsymbol{i}_3}{m} - \dot{\hat{\boldsymbol{v}}} + \boldsymbol{\varrho}_v) \tag{17}$$

where $\boldsymbol{Q}_v = \mathrm{diag}(q_{v1}, q_{v2}, q_{v3})$; $\boldsymbol{\varrho}_v = [\varrho_{v1}, \varrho_{v2}, \varrho_{v3}]^\mathsf{T}$; q_{vj} and ϱ_{vj} are given by

$$q_{vj} = \frac{\partial \varepsilon_{vj}}{\partial e_{vj}} = \frac{1}{(1+\xi_{vj})(1-\xi_{vj})}$$

$$\varrho_{vj} = \frac{\partial \xi_{vj}}{\partial t} = -\dot{\rho}_{vj}\xi_{vj}.$$

Now we are ready to design the position control law for the translation dynamic subsystem. Utilizing the transformed velocity error, the thrust magnitude is designed as

$$f = \|\boldsymbol{F}\| = \| - k_2\boldsymbol{Q}_v^\mathsf{T}\boldsymbol{\varepsilon}_v + mg\boldsymbol{i}_3\| \tag{18}$$

where $\boldsymbol{F} = -k_2\boldsymbol{Q}_v^\mathsf{T}\boldsymbol{\varepsilon}_v + mg\boldsymbol{i}_3$, and $k_2 > 0$ is a positive gain.

Remark 2: An important assumption in a classic SE(3) controller [27] is the availability of the target's velocity information for feedback. Unfortunately, this requirement is not always satisfied in reality because of the inaccurate velocity measurement of the target. What's more, the quadrotors might not be equipped with related sensors to reduce the weight and cost. Therefore, the unavailability of the target's velocity makes the landing control of quadrotors more challenging. To address the autonomous landing problem that target velocity cannot be measured, estimated, or communicated in GPS-denied environments, a low-complexity position controller is proposed without the velocity and acceleration of the target. Note the thrust force given by (18) does not incorporate any prior model knowledge of the moving platform. Furthermore, no estimations have been employed to acquire the velocity $\dot{\boldsymbol{p}}_t$ of the moving platform. Moreover, it avoids the differentiation of the virtual variable $\hat{\boldsymbol{v}}$ without using additional filters. Different from the tracking control, the landing control required a sufficiently low desired height to ensure the safe landing so that it emphasizes the transient performance of the PBVS controller. The proposed method is of significance in landing control due to it predetermining the convergence residual set, convergence rate, and overshoot of the landing errors. In short, the proposed low-complexity position controller bounds the state response of an uncertain relative dynamic system subjected to an unknown platform model, such that it is robust to an uncertain relative dynamic landing system within the bounded velocity of the moving platform.

Remark 3: The proposed PBVS method provides a low-complexity solution for nonlinear systems. The error transformation that incorporates the prescribed performance specifications is originally used in the PBVS landing control and explicitly guarantees the landing behavior, e.g., transient and steady-state performance. More significantly, with the help of the characteristics of the performance function, the designed controller avoids the explosion of complexity issues raised by traditional backstepping-like approaches. As a result, the control design is of considerably low computational complexity without utilizing additional filters.

Remark 4: A quadrotor is an aircraft that is lifted and propelled by four independent actuators. Its dynamics can be seen as the interconnection of the under-actuated position and the fully actuated attitude subsystems. Due to the highly nonlinear dynamics of the quadrotor, the landing control is addressed by the design of a low-complexity outer-loop controller and a geometric inner-loop attitude controller. In detail, the outer loop with slow dynamics generates the thrust and the desired attitude for the inner loop. Moreover, the inner loop with fast dynamics aims to track an attitude command and generate the required torques. Before starting to describe the attitude controller, we will give the desired attitude $\boldsymbol{R}_d$, which can be extracted from the intermediary control input $\boldsymbol{F}$.

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C. Geometric Attitude Control

A geometric controller for the attitude subsystem (3) and (4) is developed to follow a desired yaw angle ψ^* because the angle information of the target is unavailable. But it is sufficient to perform the landing mission due to the symmetry of the physical structure of the quadrotor. The desired attitude $\boldsymbol{R}_d$ is designed on SO(3) has the advantage of the large-angle maneuvers. Hence, the error function on SO(3) is configured to stabilize the attitude represented by an identity matrix $\boldsymbol{I}$ as

$$\Upsilon(\boldsymbol{R}, \boldsymbol{R}_d) = \frac{1}{2} tr[\boldsymbol{I} - \boldsymbol{R}_d^T \boldsymbol{R}]. \tag{19}$$

The error function Υ is locally positive-definite about $\boldsymbol{R} = \boldsymbol{R}_d$. For a given $\boldsymbol{R}_d$, this set can be represented by the sublevel set $L = \{\boldsymbol{R} \in \mathrm{SO}(3) \| \Upsilon(\boldsymbol{R}, \boldsymbol{R}_d) < 2\}$. The desired rotation matrix $\boldsymbol{R}_d = [\boldsymbol{r}_{d1}, \boldsymbol{r}_{d2}, \boldsymbol{r}_{d3}]$ is extracted from F as follows:

$$\begin{cases} \boldsymbol{r}_{d1} = \boldsymbol{r}_{d2} \times \boldsymbol{r}_{d3} \\ \boldsymbol{r}_{d2} = \frac{\boldsymbol{r}_{d3} \times \boldsymbol{a}}{\|\boldsymbol{r}_{d3} \times \boldsymbol{a}\|} \\ \boldsymbol{r}_{d3} = \frac{\boldsymbol{F}}{\|\boldsymbol{F}\|} \end{cases} \tag{20}$$

where $\boldsymbol{a} = [\cos(\psi^*), \sin(\psi^*), 0]^T$. Given $\boldsymbol{R}_d$ and $\boldsymbol{\Omega}_d$, the attitude error and angular velocity error are given by

$$\boldsymbol{e}_R = \frac{1}{2}(\boldsymbol{R}_d^{\mathsf{T}} \boldsymbol{R} - \boldsymbol{R}^{\mathsf{T}} \boldsymbol{R}_d)^{\vee} \tag{21}$$

$$\boldsymbol{e}_\Omega = \boldsymbol{\Omega} - \boldsymbol{R}^{\mathsf{T}} \boldsymbol{R}_d \boldsymbol{\Omega}_d. \tag{22}$$

Motivated by the standard geometric attitude controller in [27], the moment input $\boldsymbol{\tau}$ is designed as

$$\boldsymbol{\tau} = -k_R \boldsymbol{e}_R - k_\Omega \boldsymbol{e}_\Omega + \boldsymbol{\Omega} \times \boldsymbol{J}\boldsymbol{\Omega} - \boldsymbol{J}(\boldsymbol{\Omega}^{\vee} \boldsymbol{R}_d^{\mathsf{T}} \boldsymbol{R} \boldsymbol{\Omega}_d - \boldsymbol{R}^{\mathsf{T}} \boldsymbol{R}_d \dot{\boldsymbol{\Omega}}_d) \tag{23}$$

where the operator $(\cdot)^{\vee}$ is the inverse of the $(\cdot)^{\wedge}$ operator such that: $SO(3) \mapsto \mathbb{R}^3$. The positive constant gains k_R, k_Ω are selected to guarantee stable performance.

D. Stability Analysis

Theorem 1: Consider the system (1)–(4) with the landing control law for the quadrotor by thrust force (18) and torque inputs (23), then the landing error signals (13), (16), and (21), (22) are uniformly ultimately bounded, and the prescribed performance (8) is guaranteed.

Proof: We first consider the stability of the attitude error dynamics. Substituting (23) into (4), then the attitude error dynamics is given by

$$\dot{\Upsilon}(\boldsymbol{R}, \boldsymbol{R}_d) = \boldsymbol{e}_R^T \boldsymbol{e}_\Omega \tag{24}$$

$$\boldsymbol{J}\dot{\boldsymbol{e}}_\Omega = -k_R \boldsymbol{e}_R - k_\Omega \boldsymbol{e}_\Omega. \tag{25}$$

Consider the following Lyapunov candidate function as

$$\boldsymbol{V} = \frac{1}{2} \boldsymbol{e}_\Omega^T \boldsymbol{J} \boldsymbol{e}_\Omega + k_R \Upsilon(\boldsymbol{R}, \boldsymbol{R}_d). \tag{26}$$

Then, the time derivative of $\boldsymbol{V}$ is

$$\dot{\boldsymbol{V}} = \boldsymbol{e}_\Omega^T \boldsymbol{J} \dot{\boldsymbol{e}}_\Omega + k_R \boldsymbol{e}_R^T \boldsymbol{e}_\Omega = -k_\Omega \|\boldsymbol{e}_\Omega\|^2 \leq 0. \tag{27}$$

Since $\dot{\boldsymbol{V}}$ is semi-negative definite, we have $\boldsymbol{e}_\Omega \to 0$ when $t \to \infty$ based on the Lyapunov stability theory. Moreover, it can be seen that $\dot{\boldsymbol{e}}_\Omega$ is uniformly continuous. Applying the Barbalat's Lemma to (25), we get $\dot{\boldsymbol{e}}_\Omega \to 0$ as $t \to \infty$. therefore $\boldsymbol{e}_R \to 0$ when $t \to \infty$.

Next, the stability of the translational error dynamics is presented. Substituting (15) into (14), then the time derivative of $\boldsymbol{\varepsilon}_p$ is given by

$$\dot{\boldsymbol{\varepsilon}}_p = -k_1 \boldsymbol{Q}_p \boldsymbol{Q}_p^{\mathsf{T}} \boldsymbol{\varepsilon}_p + \boldsymbol{Q}_p(\boldsymbol{\varrho}_p + \boldsymbol{e}_v - \dot{\boldsymbol{p}}_t) \tag{28}$$

since $\boldsymbol{R}_d \boldsymbol{i}_3 = \boldsymbol{r}_{d3} = \boldsymbol{F}/\|\boldsymbol{F}\|$, the time derivative of velocity error dynamics $\boldsymbol{\varepsilon}_v$ can be obtained as

$$\dot{\boldsymbol{\varepsilon}}_v = \boldsymbol{Q}_v(-g i_3 + \frac{\boldsymbol{F}}{m} + \boldsymbol{\varrho}_v - \dot{\hat{\boldsymbol{v}}} + \boldsymbol{\chi}) \tag{29}$$

where $\boldsymbol{\chi} = \frac{f}{m}(\boldsymbol{R}\boldsymbol{i}_3 - \boldsymbol{R}_d \boldsymbol{i}_3)$.

To prove the stability of the position controller, we should show the boundedness of the transformed error ε_i. Owing to the properties of the T-function appearing in (10), it leads to proving that $\xi_i = e_i/\rho_i$ evolves strictly within the open set $(-1, 1)$. Therefore, we can further demonstrate all errors achieve the predetermined performance specified by (8) as $-\rho_i(t) < e_i(t) < \rho_i(t)$. In the following, we proceed in three steps to prove the stability of the position control law. In step 1, the existence and uniqueness of a maximal system solution $\boldsymbol{\xi}_j : [0, \tau_{\max}) \to \boldsymbol{A}_\xi$ for some $\tau_{\max} > 0$, is guaranteed. In step 2, the boundedness of all closed-loop signals is proven, and their boundaries belong to a compact subset of $\boldsymbol{A}_\xi$. Consequently, in step 3, the points raised in the second step are proven to remain when $\tau_{\max} \to \infty$.

Step 1: Prove the existence and uniqueness of a maximal solution over a time interval $[0, \tau_{\max})$. We first defined the open set as follows:

$$\boldsymbol{A}_\xi = (-1, 1) \times (-1, 1). \tag{30}$$

The set $\boldsymbol{A}_\xi$ is nonempty and open. Using (8) and (9), since the constructed performance function satisfies $\|\rho_{pj}(0)\| > \|e_{pj}(0)\|$ and $\|\rho_{vj}(0)\| > \|e_{vj}(0)\|$, therefore, the $\|\xi_{pj}(0)\| = \|e_{pj}(0)/\rho_{pj}(0)\| < 1$ and $\|\xi_{vj}(0)\| = \|e_{vj}(0)/\rho_{vj}(0)\| < 1$. If we define $\boldsymbol{\xi}_j = [\xi_{pj}, \xi_{vj}]^T$, where $j = 1, 2, 3$. As a result, the $\boldsymbol{\xi}_j \in \boldsymbol{A}_\xi$. The time derivative of $\boldsymbol{\xi}_j$ is given by

$$\dot{\boldsymbol{\xi}}_j = \begin{bmatrix} \frac{1}{\rho_{pj}}(\frac{\dot{e}_{pj}}{\rho_{pj}} - \xi_{pj}\dot{\rho}_{pj}) \\ \frac{1}{\rho_{vj}}(\frac{\dot{e}_{vj}}{\rho_{vj}} - \xi_{vj}\dot{\rho}_{vj}) \end{bmatrix}. \tag{31}$$

Since the fact that the performance functions ρ_{pj} and ρ_{vj} are bounded, the $\dot{\rho}_{pj}, \xi_{pj}, \dot{\rho}_{vj}$, and ξ_{vj} are bounded and continuously differentiable functions of time. Utilizing (15) and (18), the intermediate control signal $\hat{\boldsymbol{v}}$ and control law f are smooth over $\boldsymbol{A}_\xi$, then $\dot{e}_{pj}$ and $\dot{e}_{vj}$ are continuously differentiable functions. Therefore, $\dot{\boldsymbol{\xi}}_j$ is bounded and continuously differentiable as well as locally Lipschitz $\boldsymbol{\xi}_j$ over the set $\boldsymbol{A}_\xi$. Thus, a uniqueness of a maximal solution $\boldsymbol{\xi}_j : [0, \tau_{\max}) \to \boldsymbol{A}_\xi$ of $\dot{\boldsymbol{\xi}}_j$ over the set $\boldsymbol{A}_\xi$ for a time interval $[0, \tau_{\max})$ is guaranteed, which implies

$$\xi_{pj} \in (-1, 1), \tilde{}\xi_{vj} \in (-1, 1), \ \forall\, t \in [0, \tau_{\max}). \tag{32}$$

Step 2: The boundedness of all closed-loop signals is proven, and their boundaries belongs to a compact subset of $\boldsymbol{A}_\xi$ for all

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$t \in [0, \tau_{\max})$. We define the second Lyapunov function as

$$\boldsymbol{V}_1 = \frac{1}{2}\boldsymbol{\varepsilon}_p^T \boldsymbol{\varepsilon}_p. \tag{33}$$

The time derivative of $\boldsymbol{V}_2$ is given by

$$\dot{\boldsymbol{V}}_1 = -k_1 \boldsymbol{\varepsilon}_p^T \boldsymbol{Q}_p \boldsymbol{Q}_p^{\mathsf{T}} \boldsymbol{\varepsilon}_p + \boldsymbol{\varepsilon}_p^T \boldsymbol{Q}_p \boldsymbol{\varpi}_1 \tag{34}$$

where $\boldsymbol{\varpi}_1 = \boldsymbol{\varrho}_p + \boldsymbol{e}_v - \dot{\boldsymbol{p}}_t$. Since the fact that selected performance function ρ_{pj}, $\dot{\rho}_{pj}$, ρ_{vj}, and $\dot{\rho}_{vj}$ are bound, using (32), we deduce that the boundedness of $\varrho_{pj} = -\dot{\rho}_{pj}\xi_{pj}$ and $e_{vj} = \rho_{vj}\xi_{vj}$ for all $t \in [0, \tau_{\max})$. Due to $\|\dot{\boldsymbol{p}}_t\| < \bar{\nu}_t$ is bounded, so there exists a constant $\Delta_1 > 0$ such that

$$\|\boldsymbol{\varpi}_1\| \leq \|\boldsymbol{\varrho}_p\| + \|\boldsymbol{e}_v\| + \|\dot{\boldsymbol{p}}_t\| \leq \Delta_1 \tag{35}$$

using Young's inequality, one has

$$\boldsymbol{\varepsilon}_p^T \boldsymbol{Q}_p \boldsymbol{\varpi}_1 \leq \sigma_1 \|\boldsymbol{Q}_p^T \boldsymbol{\varepsilon}_p\|^2 + \frac{1}{4\sigma_1}\|\boldsymbol{\varpi}_1\|^2 \tag{36}$$

where σ_1 is a positive constant. Then, (34) can be rewritten as

$$\dot{\boldsymbol{V}}_1 \leq -c_1 \boldsymbol{V}_1 + b_1 \tag{37}$$

where $c_1 = (k_1 - \sigma_1)\lambda_{\min}^2(\boldsymbol{Q}_p)$, and $b_1 = \Delta_1^2/4\sigma_1$. Since the normalized state error $\xi_{pj} \in (-1, 1)$, then $q_{pj} = 1/(1 - \xi_{vj}^2) \geq 1$ is bounded, so $\lambda_{\min}(\boldsymbol{Q}_p) \geq 1, \forall t \in [0, \tau_{\max})$. we assume that k_1 and σ_1 are chosen such that $k_1 \geq \sigma_1$, then $c_1 > 0$. Using the Lyapunov stability theory, it can be concluded that the boundedness of $\boldsymbol{\varepsilon}_p$ such that $\|\varepsilon_{pj}\| < \bar{\varepsilon}_{pj}$, where $j = 1, 2, 3$. Furthermore, taking the inverse logarithmic function for (12), we get

$$-1 < \frac{e^{-2\bar{\varepsilon}_{pj}} - 1}{e^{-2\bar{\varepsilon}_{pj}} + 1} \leq \xi_{pj} \leq \frac{e^{2\bar{\varepsilon}_{pj}} - 1}{e^{2\bar{\varepsilon}_{pj}} + 1} < 1 \tag{38}$$

for all $t \in [0, \tau_{\max})$. The time derivative of $\hat{\boldsymbol{v}}$ is

$$\dot{\hat{\boldsymbol{v}}} = -k_1 \dot{\boldsymbol{Q}}_p^{\mathsf{T}} \boldsymbol{\varepsilon}_p - k_1 \boldsymbol{Q}_p^{\mathsf{T}} \dot{\boldsymbol{\varepsilon}}_p \tag{39}$$

utilizing (38), it guarantees the boundedness of $\boldsymbol{Q}_p$ and $\dot{\boldsymbol{Q}}_p$, $\dot{\boldsymbol{\varepsilon}}_p$ for all $t \in [0, \tau_{\max})$. Hence, we obtain the boundedness of $\dot{\hat{\boldsymbol{v}}}$, $\forall t \in [0, \tau_{\max})$.

We take the third Lyapunov function as

$$\boldsymbol{V}_2 = \frac{1}{2}\boldsymbol{\varepsilon}_v^T \boldsymbol{\varepsilon}_v. \tag{40}$$

The time derivative of $\boldsymbol{V}_2$ is given by

$$\dot{\boldsymbol{V}}_2 = -\frac{k_2}{m}\boldsymbol{\varepsilon}_v^T \boldsymbol{Q}_v \boldsymbol{Q}_v^{\mathsf{T}} \boldsymbol{\varepsilon}_v + \boldsymbol{\varepsilon}_v^T \boldsymbol{Q}_v \boldsymbol{\varpi}_2 + \boldsymbol{\varepsilon}_v^T \boldsymbol{Q}_v \boldsymbol{\chi} \tag{41}$$

where $\boldsymbol{\varpi}_2 = \boldsymbol{\varrho}_v - \dot{\hat{\boldsymbol{v}}}$. Since the selected performance function ρ_{vj}, $\dot{\rho}_{vj}$ are bound, using (32), we deduce that the boundedness of $\varrho_{vj} = -\dot{\rho}_{vj}\xi_{vj}$ for all $t \in [0, \tau_{\max})$. Since the $\dot{\hat{\boldsymbol{v}}}$ is bounded, there exists a constant $\Delta_2 > 0$ such that

$$\|\boldsymbol{\varpi}_2\| \leq \|\boldsymbol{\varrho}_v\| + \|\dot{\hat{\boldsymbol{v}}}\| \leq \Delta_2. \tag{42}$$

Moreover, according to Rodrigues' formula, we obtain $\|\boldsymbol{e}_R\| = \sin\|x\|$, when $\boldsymbol{R}_d^T \boldsymbol{R} = \exp \hat{x}$ for $x \in \mathbb{R}^3$. This represents the sine of the eigen-axis rotation angle between $\boldsymbol{R}$ and $\boldsymbol{R}_d$ such that

$$\|(\boldsymbol{R}\boldsymbol{i}_3 - \boldsymbol{R}_d \boldsymbol{i}_3)\| \leq \|\boldsymbol{e}_R\| = \beta < 1. \tag{43}$$

The last term of the right-hand side of (41) is bounded as

$$\|\boldsymbol{\varepsilon}_v^T \boldsymbol{Q}_v \boldsymbol{\chi}\| \leq \left[k_2 \|\boldsymbol{Q}_v^{\mathsf{T}} \boldsymbol{\varepsilon}_v\|^2 + \|\boldsymbol{Q}_v^{\boldsymbol{T}} \boldsymbol{\varepsilon}_v\| \|mg\boldsymbol{i}_3\|\right] \frac{\beta}{m}. \tag{44}$$

Using inequalities (42), (44), and applying Young's inequality, then (41) yields

$$\begin{aligned} \dot{\boldsymbol{V}}_2 &\leq -\frac{k_2(1-\beta)}{m}\boldsymbol{\varepsilon}_v^T \boldsymbol{Q}_v \boldsymbol{Q}_v^{\mathsf{T}} \boldsymbol{\varepsilon}_v + (\Delta_2 + g)\|\boldsymbol{Q}_v^T \boldsymbol{\varepsilon}_v\| \\ &\leq -\left(\frac{k_2(1-\beta)}{m} - \sigma_2\right)\|\boldsymbol{Q}_v^T \boldsymbol{\varepsilon}_v\|^2 + \frac{(\Delta_2 + g)^2}{4\sigma_2} \end{aligned} \tag{45}$$

where σ_2 is a positive constant. Inequality (45) implies that

$$\dot{\boldsymbol{V}}_2 \leq -c_2 \boldsymbol{V}_2 + b_2 \tag{46}$$

where $c_2 = (k_2(1-\beta)/m - \sigma_2)\lambda_{\min}^2(\boldsymbol{Q}_v)$, and $b_2 = \frac{(\Delta_2+g)^2}{4\sigma_2}$. Since the normalized state error $\xi_{vj} \in (-1, 1)$, $q_{vj} = 1/(1 - \xi_{vj}^2) \geq 1$, and, therefore, $\lambda_{\min}(\boldsymbol{Q}_v) \geq 1$. We assume that k_2 and σ_2 are chosen such that $k_2(1-\beta)/m > \sigma_2$ and hence $c_2 > 0$. Therefore $\boldsymbol{\varepsilon}_v$ is bounded such that $\|\varepsilon_{vj}\| < \bar{\varepsilon}_{vj}$, where $j = 1, 2, 3$. Taking the inverse logarithmic function for (12), for all $t \in [0, \tau_{\max})$, we obtain

$$-1 < \frac{e^{-2\bar{\varepsilon}_{vj}} - 1}{e^{-2\bar{\varepsilon}_{vj}} + 1} \leq \xi_{vj} \leq \frac{e^{2\bar{\varepsilon}_{vj}} - 1}{e^{2\bar{\varepsilon}_{vj}} + 1} < 1. \tag{47}$$

The boundedness of all closed-loop signals of $\boldsymbol{\xi}_j$ evolves in a compact subset of $\boldsymbol{A}_\xi$, so the errors are uniformly ultimately bounded for all $t \in [0, \tau_{\max})$.

Step 3: The points raised in the second step are proven to remain when $\tau_{\max} \to \infty$. Using (38) and (47), It's verified that $\boldsymbol{\xi}_j(t) \in \boldsymbol{A}'_\xi$ for all $t \in [0, \tau_{\max})$, where the set $\boldsymbol{A}'_\xi = (\underline{\xi}_{pj}, \bar{\xi}_{pj}) \times (\underline{\xi}_{vj}, \bar{\xi}_{vj}) \in \boldsymbol{A}_\xi$ is nonempty and compact. However, [28] dictates the existence of a time instant $t' \in [0, \tau_{\max})$ such that $\boldsymbol{\xi}_j(t') \notin \boldsymbol{A}'_\xi$. Thus, $\tau_{\max} = \infty$ by contradiction. This completes the proof.

E. Modified Attitude Control Based on a Command Filter

To avoid complex calculations in the torque input, a command filter is introduced to compute the $\dot{\boldsymbol{F}}$ without the differentiation so that $\dot{\boldsymbol{R}}_d$ can be easily obtained. Based on (20), $\dot{\boldsymbol{R}}_d = [\dot{\boldsymbol{r}}_{d1}, \dot{\boldsymbol{r}}_{d2}, \dot{\boldsymbol{r}}_{d3}]$ as the derivative of $\boldsymbol{R}_d$ is given by

$$\begin{cases} \dot{\boldsymbol{r}}_{d1} = \dot{\boldsymbol{r}}_{d2} \times \boldsymbol{r}_{d3} + \boldsymbol{r}_{d2} \times \dot{\boldsymbol{r}}_{d3} \\ \dot{\boldsymbol{r}}_{d2} = (\boldsymbol{I}_{3\times3} - \boldsymbol{r}_{d2}\boldsymbol{r}_{d2}^{\mathsf{T}})\frac{(\dot{\boldsymbol{r}}_{d3}\times\boldsymbol{a} + \boldsymbol{r}_{d3}\times\dot{\boldsymbol{a}})}{\|\boldsymbol{r}_{d3}\times\boldsymbol{a}\|} \\ \dot{\boldsymbol{r}}_{d3} = (\boldsymbol{I}_{3\times3} - \boldsymbol{r}_{d3}\boldsymbol{r}_{d3}^{\mathsf{T}})\frac{\dot{\boldsymbol{F}}}{\|\boldsymbol{F}\|} \end{cases} \tag{48}$$

where $\boldsymbol{I}_{3\times3}$ is the identity matrix. Introducing the command filter technology [29], one can be expressed as follows:

$$\begin{cases} \dot{x}_{i,1} = \omega_n x_{i,2} \\ \dot{x}_{i,2} = -2\zeta\omega_n x_{i,2} - \omega_n(x_{i,1} - F_i) \end{cases} \tag{49}$$

with $i \in 1, 2, 3$, and the outputs of command filter are $x_{i,1} = F_i$ and $x_{i,2} = \dot{F}_i$. Filter parameters are designed as $\omega_n > 0$ and $\zeta \in (0, 1]$. We design the modified attitude controller as

$$\boldsymbol{\tau} = -K_R \boldsymbol{e}_R - K_\Omega \boldsymbol{e}_\Omega + \boldsymbol{\Omega} \times \boldsymbol{J}\boldsymbol{\Omega}. \tag{50}$$

Remark 5: Compared with the geometric controller in (23), the modified geometric attitude controller (50) neglects the last

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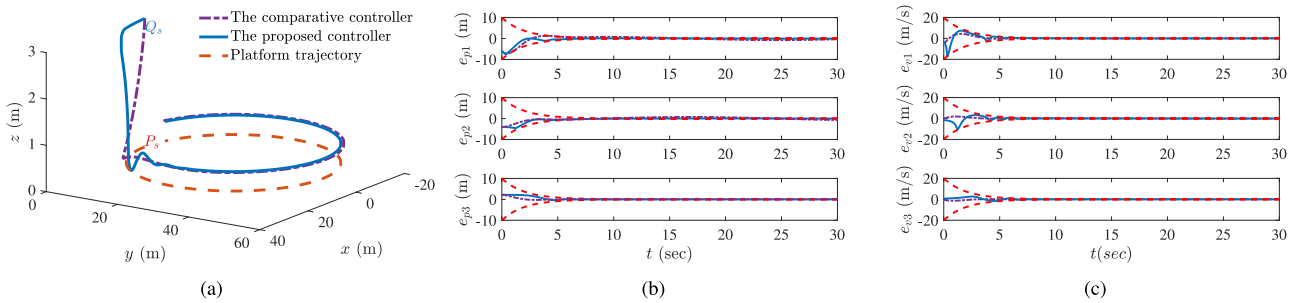

Fig. 2. Simulation results for the first comparative example. (a) Trajectories of the quadrotor and the moving platform. (b) and (c) Position errors and linear velocity errors of the proposed method (blue) along with corresponding performance functions (red) and comparative method (purple).

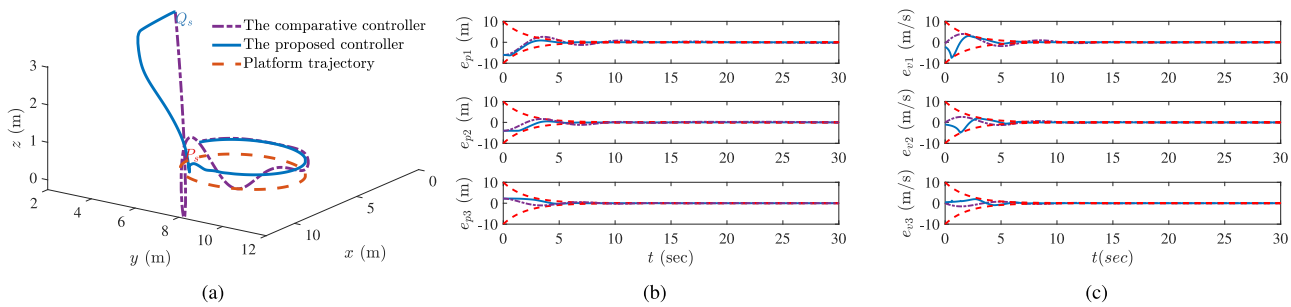

Fig. 3. Simulation results for the second comparative example. (a) Trajectories of a quadrotor and a moving platform. (b) and (c) Position errors and linear velocity errors of the proposed method (blue) along with corresponding performance functions (red) and comparative method (purple).

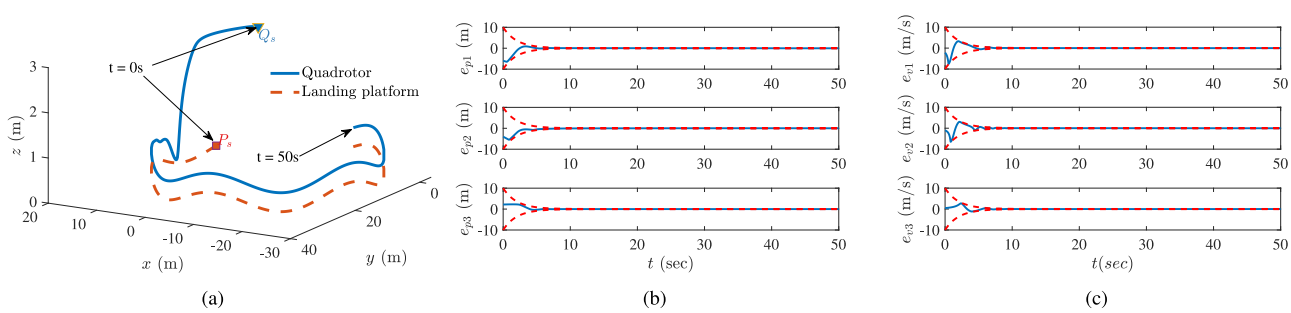

Fig. 4. Simulation results for a sinusoidally oscillating platform whose horizontal trajectory is a circle. (a) The trajectories of a quadrotor and a moving platform. (b) The time response of position errors. (c) The time response of linear velocity errors.

two terms in the controller (23). With the help of the higher order inertial cancelation terms, this controller in (23) is guaranteed to be exponentially stable for almost any rotation. These high-order terms are related to $\boldsymbol{R}_d$ and $\boldsymbol{\Omega}_d$, and they are usually small enough that we can consider them as bounded disturbances. Based on the Lyapunov stability theory, the controller in (50) can make errors converge to a small neighborhood of zero, and so it is stable in the absence of high-order terms. Also, it has been mentioned in [30] that the effects of these high-order terms are insignificant. From a practical standpoint, a lot of experiments have achieved satisfactory performance by neglecting the higher order terms.

IV. Simulation and Experimental Studies

To verify the effectiveness of the proposed control scheme, several simulations and experiments were presented in Figs. 4–8. All results were based on the following configuration. The system parameters of quadrotor were listed as: $m = 2.3$ kg, $g = 9.8$ m/s^2 and $\boldsymbol{J} = \mathrm{diag}(0.0432, 0.0432, 0.0715)$ kg·m^2/rad^2. It can be seen from the stability proof that the controller gains are chosen such that $k_\Omega > 0$, $k_1 \geq \sigma_1$ and $k_2(1-\beta)/m > \sigma_2$. Since $\beta < 1$, $m > 0$, and σ_1, σ_2 are arbitrarily small positive constants, the control gains are positive and obtained via a trial and error approach when satisfactory performance is achieved. The parameters of the controllers were set as $k_1 = 4.5$, $k_2 = 2.0$, $k_R = 150$ and $k_\Omega = 0.4$, and the gains of the filter were chosen as $\zeta = 0.5, \omega_n = 0.02$. The parameters of the performance function were given by $\rho_{pj}^\infty = 0.05$ m, $\rho_{vj}^\infty = 0.05$ m/s and $l_{pj} = l_{vj} = 0.5$, where $j = 1, 2, 3$. In addition, an Extend Kalman Filter was adopted to integrate the visual information and IMU to calculate a more reliable relative position. The desired relative position was set as $\boldsymbol{r}_{pt}^* = (0, 0, 0.4)$ m. Before investigating experimental results by using hardware, the simulation results allow us to gain insights into the proposed method in different

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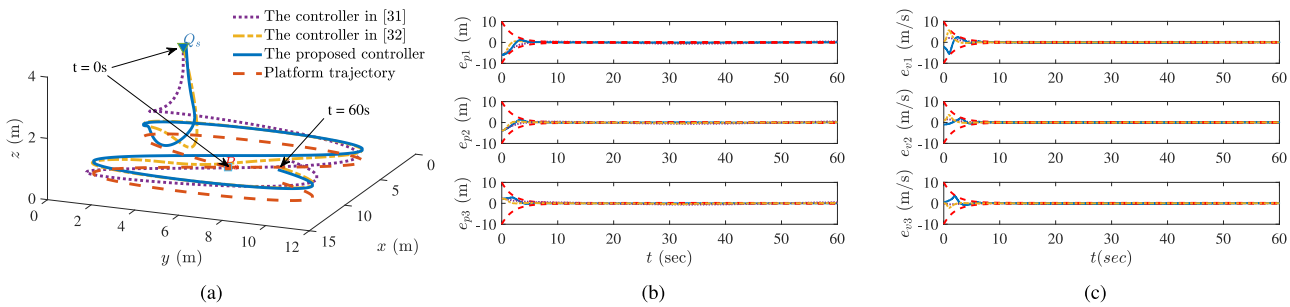

Fig. 5. Simulation results with a target having the "8" shaped trajectory, where the measurement noises and the wind disturbances are imposed at the side of the quadrotor. (a) The trajectories of a quadrotor and a moving platform, where the purple dotted line is the response curve of controller [31], the yellow dot-and-dash line is the response curve of controller [32], the blue solid line is the proposed controller, and the red dashed line is the trajectory of the target. (b) The time response of position errors. (c) The time response of linear velocity errors.

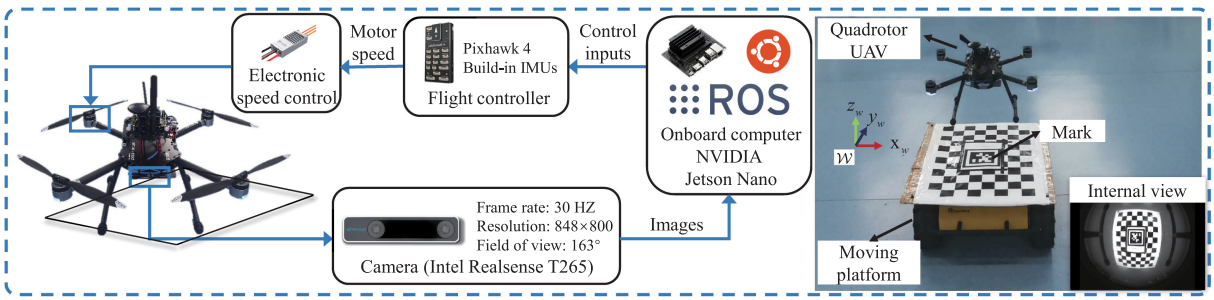

Fig. 6. Experimental setup for the vision-based autonomous landing system, and the internal view from the embedded camera.

scenarios of interest. Hence, these results may be of a larger scale than they are possible in the hardware setup, e.g., due to restrictions regarding laboratory size.

A. Simulation Results Considering Two Comparative Studies

In this section, two comparative examples were conducted to verify the effectiveness of the proposed controller. In the first example, we evaluated the performance of the proposed controller by comparing it with a geometric controller [30] with the target's velocity feedback. The linear and angular velocities of the target were set as 5 m/s and 0.2 rad/s, respectively. The results are depicted in Fig. 2. Fig. 2(a) shows the trajectories of the quadrotor and the target. Though the geometric controller with the target's velocity feedback converges faster, both of their trajectories are smooth and have good performance. The time response of position and velocity errors are depicted in Fig. 2(b) and (c). It can be observed that all errors of the proposed method are always guaranteed within the predefined performance bounds, so that it has a smaller convergence residual set. It is important for landing missions with strict accuracy requirements.

In the second example, under the same simulation conditions, the comparative controller in [30] was implemented without the target's velocity. To ensure the comparative method performs as well as possible in this scenario, the linear and angular velocities of the landing platform were set as 0.5 m/s and 0.2 rad/s. The results are shown in Fig. 3. Fig. 3(a) illustrates the trajectories of the quadrotor and the target. It can be easily seen that the trajectory of the proposed controller is smoother, while the comparative controller oscillates greatly. In Fig. 3(b) and (c), the time response of position and velocity errors were presented. From the perspective of transient control performance, the proposed method has a faster convergence speed and smaller overshoot. As far as the actual landing mission is concerned, the quadrotor is not allowed to have too much overshoot to cause a collision during the landing. Obviously, the comparison method is not up to this requirement when the target's velocity information is missing. More significantly, the proposed method is adapted to the complex landing environment due to the fact that it can freely select the performance boundary. Therefore, the comparative simulation studies verify the effectiveness of the proposed method without the velocity measurements of the target.

B. Simulation Results for Targets Moving in 3-D Space

Based on the horizontal motions of the target, the vertical motions have been added to demonstrate the effectiveness of the landing controller for a target moving in a 3-D space. The linear and angular velocities of the landing platform were set as 2 m/s and 0.2 rd/s. The chosen vertical reference trajectory can be described as $z_r = 0.2\sin(0.5t)$. The simulation results are shown in Fig. 4. Fig. 4(a) describes the X–Y–Z positions of the quadrotor and the target. The time responses of position and linear velocity errors are plotted in Fig. 4(b) and (c). It can be seen that all errors are always guaranteed within the predefined performance bounds. Hence, the quadrotor can track the reference trajectory effectively though the target has one more degree of freedom in the vertical axis. Therefore, the proposed control method can achieve fast and accurate landing

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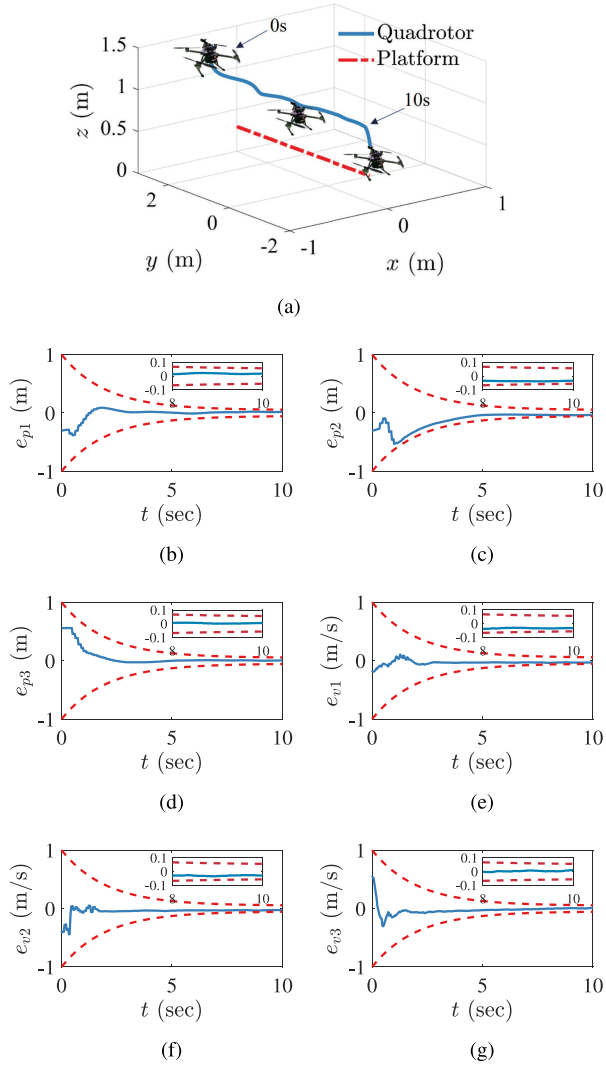

Fig. 7. Experiment 1: Quadrotor lands on a platform moving along a straight line: (a) Trajectories of a quadrotor and a moving platform for the whole landing process. (b)–(g) The position and linear velocity errors (blue) and the corresponding imposed performance bounds (red).

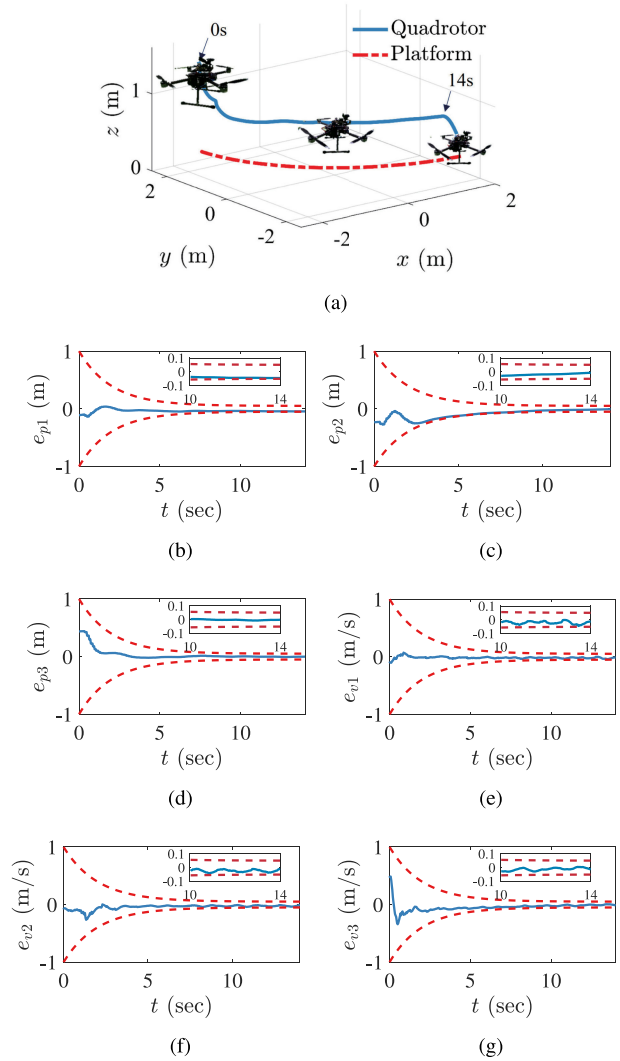

Fig. 8. Experiment 2: Quadrotor lands on a platform moving along a circular trajectory: (a) Trajectories of a quadrotor and a moving platform for the whole landing process. (b)–(g) The position and linear velocity errors (blue) and the corresponding imposed performance bounds (red).

of the quadrotor on the target moving in 3-D space with the designed transient and steady-state performance indicators.

C. Simulation Results for Strong Robustness Verification

In this simulation, the performance of the proposed controller was evaluated for the real application scenarios, such as the uncertain target with complex motions, wind disturbances, and measurement noises. To fit the real situation, we considered that the wind caused a time-varying accelerated speed disturbance as $0.2\cos(0.1t)$ m/s^2 to the x-axis, y-axis, and z-axis in the Cartesian space; the noise was imposed on the measurements with a standard deviation of 0.015 m; the target was endowed with complex motions involving acceleration and deceleration, and an "8″ shaped reference trajectory was chosen as $x_r = 5\sin(0.1t)$, $y_r = 5\sin(0.2t)$. In addition, the following comparison numerical simulations have been conducted to further demonstrate the strong robustness of the proposed controller. Under the same simulation conditions, we compared the results with those of the approaches based on an adaptive backstepping controller in [31] and based on a boundary layer sliding controller in [32]. The control schemes in [31] and [32] are similarly defined for the system modeled by (1)–(4), but they both require the velocity and acceleration of the target. Besides, the controller in [32] assumed that the total wind speed relative to the UAV is known to estimate the acceleration caused by drag. The results are shown in Fig. 5. Fig. 5(a) depicts 3-D trajectories of the quadrotor and the target. The time responses of the position and linear velocity errors are plotted in Fig. 5(b) and (c). From Fig. 5(a), the trajectory of the controller in [31] is not satisfactory since it cannot perfectly compensate for the influence of the external wind disturbance. The trajectory quality of the controller in [32] has been modified with the help of the observer. In contrast, the proposed controller can guarantee the best trajectory according to the prescribed transient and steady-state performance although in the absence of the target's velocity and the disturbance observers. It can be seen that the frequent variations of target velocities arose certain tracking errors and the wind caused a time-varying accelerated speed disturbance. However, all errors of the proposed controller

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not only converged quickly but also strictly retained within the prescribed performance bounds. These studies show the superior robustness of the proposed approach over existing controllers in handling model uncertainties and disturbances.

D. Experimental Results

Two real-hardware experiments were presented to further demonstrate the performance of a proposed control scheme on a physical quadrotor P600. As shown in Fig. 6, the experimental setup of quadrotor included a downward-looking wide-angle camera Realsense T265 with a 163° field of view (FOV) only used to get RGB images, a custom-made electronics flight controller board Pixhawk 4 (build-in IMUs) provided angle and velocity information of quadrotor, and a jetson nano onboard computer ran the closed-loop control system at 20 Hz. The proposed algorithm was implemented in the robot operating system kinetic. A motion capture system was only utilized to illustrate the landing trajectories. Two cases of experiments were considered as follows:

Case 1: Quadrotor landed on a platform moving along a straight line.

Case 2: Quadrotor landed on a platform moving along a circular trajectory.

In the first case, the linear velocity of the platform was set as 0.5 m/s. The results are shown in Fig. 7, Fig. 7(a) depicts the 3-D trajectories of the quadrotor and the moving platform for the whole landing process. In Fig. 7(b)–(d), it can be seen that the position errors converge with a satisfactory accuracy range of about 0.035 m. The linear velocity errors remain within the predefined region and ensure that they will never violate the prescribed transient bounds in Fig. 7(e)–(g). The results show the good stability and performance of the proposed control scheme for landing on a moving platform.

In the second case, the performance of the landing controller with nonlinear reference velocity was evaluated. The linear and angular velocities of the Husky moving platform were given by 0.5 m/s and 0.2 rd/s. Experimental results were illustrated in Fig. 8. Fig. 8(a) illustrates the 3-D trajectories of the quadrotor and the moving platform. One can see that position and linear velocity errors are regulated to a predefined arbitrarily small residual set along with the absolute decaying time function in Fig. 8(b)–(g). The results show that the transient and steady-state landing behaviors of the proposed controller are explicitly guaranteed and verify the robustness of the proposed controller against the uncertain relative dynamic system subjected to unknown moving platform models.

V. Conclusion

This article addressed the PBVS landing control problem of an underactuated quadrotor on an unknown model of moving platform. The proposed controller is of considerably low computational complexity, so it is suitable for this type of vehicle with low cost, real-time operation, and limited computation ability. Under the hierarchical control framework involving a position controller and an attitude controller, the quadrotor underactuation was well treated. More significantly, the PBVS controller using the error transformation technology and the geometric attitude controller based on the filter were designed to overcome the lack of the target's velocity. Therefore, the landing controller exhibited strong robustness against the uncertain relative dynamic system subjected to an unknown model of moving platform. With this method, the transient and steady-state landing behaviors were explicitly guaranteed. Hence, the proposed controller could meet the landing requirements for many applications of interest. Numerical simulations and experiments were conducted to show the validity and robustness of the proposed approach. Future work is to improve the performance of our landing algorithm for considering the FOV constraints of cameras.

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